

Integrated Data Analytics Project Report

A Complete, Deployable Cross-Sell Propensity System for FirstBank Nigeria Limited

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1. Business Process Identification and Problem Definition

FirstBank Nigeria Limited, the oldest commercial bank in West Africa, serves over 42 million retail customer accounts through more than 820 business offices (First Bank of Nigeria, 2024). Despite this scale, relationship managers and marketing teams rely on broad demographic proxies, age and branch location, to decide which customers receive cross-sell offers, rather than how customers actually use their accounts. This produces generic offers, wasted marketing spend, and missed opportunities to deepen relationships with the right customers (Module 1 Vision Document).

This is a strategically material problem, not a cosmetic one. FirstBank has publicly signaled a shift toward the middle-class consumer market and diversified retail portfolios, since consumer spending remains a comparatively small share of Nigeria's GDP relative to developed markets (Marketing Edge Magazine, 2024). Undifferentiated marketing directly undercuts that strategy: a bank that cannot tell which customers are ready for a new product cannot execute a diversification strategy efficiently. The measurable objective set in Module 1, a 15 percent uplift in cross-sell conversion within two quarters, alongside identification of at least three underserved segments for financial inclusion, has anchored every subsequent module without revision.

Ethical AI Charter

The Module 1 Vision Document committed this project to three principles that hold across all five modules and this integration: fairness (segments and predictions are evaluated for disparate impact before any commercial action), transparency (model logic is explainable in business terms to relationship managers and customers), and accountability (every automated decision affecting a customer's product access has a named human owner and an audit trail). The full charter, and how each module operationalized it, is consolidated in docs/ethical_ai_framework.md and summarized in Section 8.

These commitments were not treated as a compliance formality to satisfy once and forget. Each module was required to show its work against the charter: Module 3's pipeline logs every transformation to a privacy audit log, Module 4's fairness audit produced a concrete correction rather than a passing mention, and this integration adds a monitoring script that keeps checking after deployment rather than only at handoff. The strategic relevance is direct: a bank whose analytics program cannot demonstrate fairness and auditability to a regulator carries reputational and regulatory risk that a 15 percent conversion uplift would not offset.

2. Project Overview and Documentation

The Module 2 Project Overview Report translated the Module 1 vision into an executable plan: a customer segmentation and propensity-scoring product, built in two-week Scrum sprints, with agile ceremonies keeping marketing and compliance feedback in the loop each cycle (Schwaber & Sutherland, 2020). A Fishbone root-cause analysis traced the low-conversion problem to four contributing causes, fragmented data, manual targeting, no segmentation model, and relationship managers' reliance on intuition, each linked to a measurable KPI carried through the rest of the project.

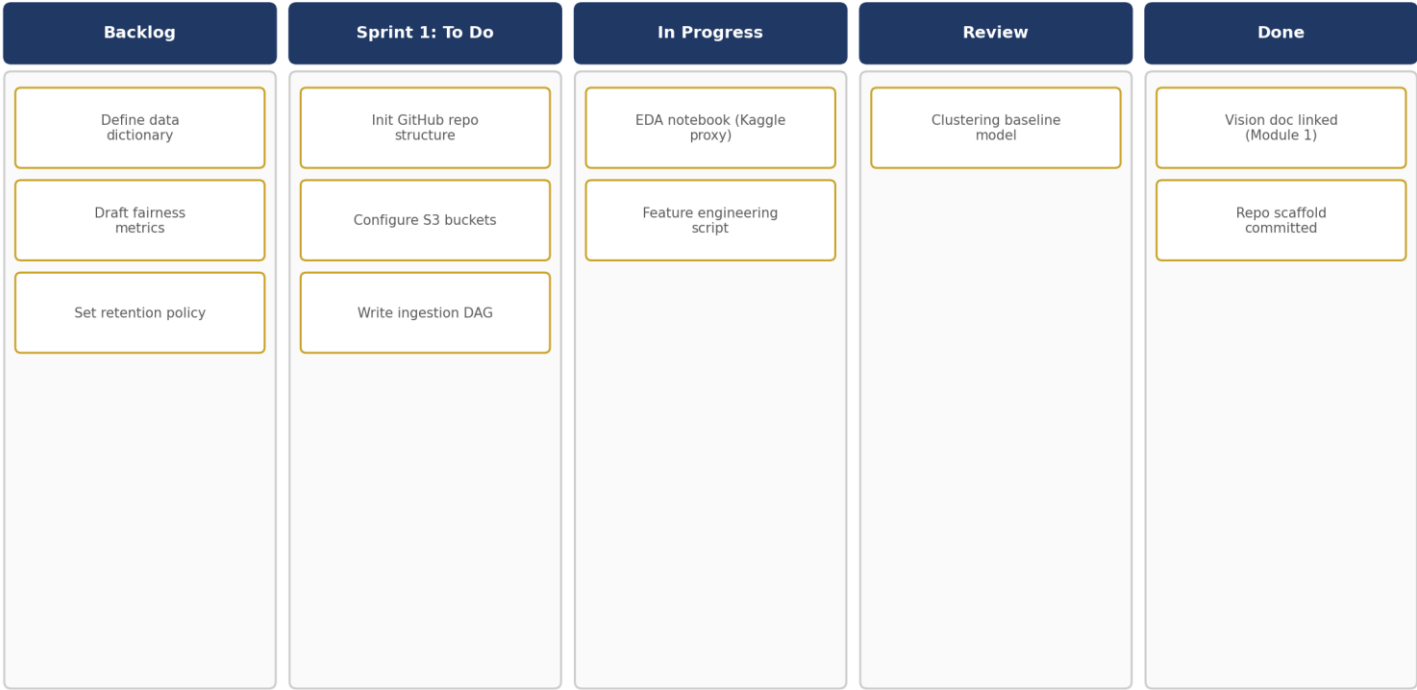
Item	Detail
Objectives	+15% cross-sell conversion, 3+ underserved segments identified, 85% quarter-over-quarter segment stability, quarterly fairness audit
Data sources	Kaggle Bank Customer Segmentation dataset (Bansal, 2021), used as a structurally realistic proxy for FirstBank's core banking data, which is not accessible for this academic project
Timeline	Five two-week sprints (Modules 1-5), integrated in this Final Project

Stakeholders	Head of Retail Banking, Chief Marketing Officer, Chief Data and Analytics Officer, Data Protection Officer, AI & Data Ethics Committee
GitHub Repository	github.com/OfiIreg/FirstBank-customer-segmentation , all code from Modules 2-5 and this integration

Milestones and Project Board

The GitHub Project Board (Kanban) tracked sprint work across Backlog, Sprint To Do, In Progress, Review, and Done columns, with tickets mapped directly to the repository structure proposed in Module 1, for example a ticket to build the ingestion DAG mapped to src/pipeline (Module 2 Report, Section 4.1). Sprint 1 delivered the repository scaffold and vision document link; Sprints 2 through 5 delivered the pipeline, model, dashboard, and this integration in sequence, each building only on artifacts the prior sprint had actually produced, not on planned but undelivered work.

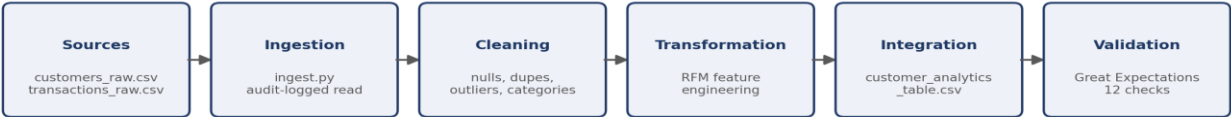
Figure 3. GitHub Project Board (Kanban) Sprint 1 Snapshot



3. Data Pipeline Design

The Module 3 pipeline moves data through seven stages, ingest, clean, transform, integrate, validate, bias-check, anonymize, orchestrated end to end by a Prefect DAG that has been run and verified, not just diagrammed.

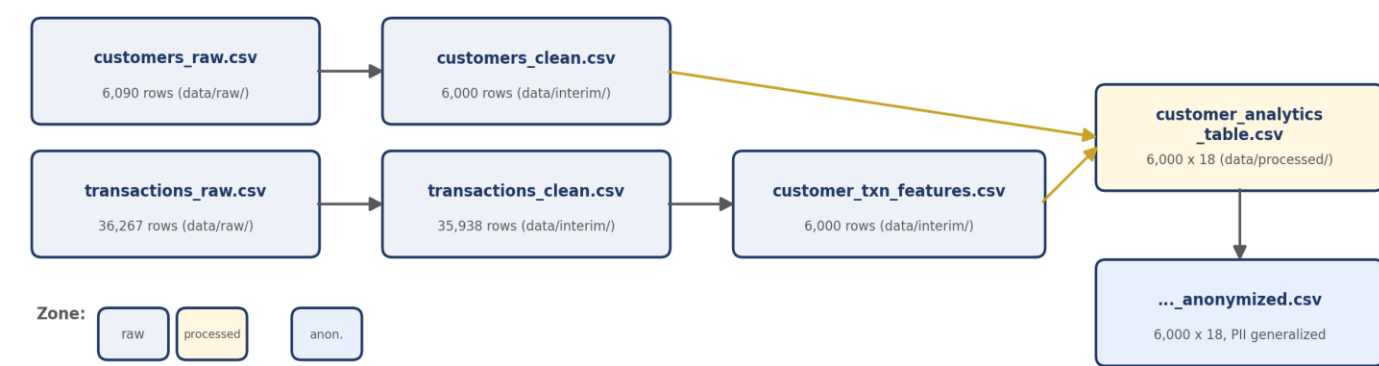
Figure 1. Data Pipeline Stages: Source to Validated Output



Data pipeline stages, from raw source to validated output (Module 3).

Cleaning resolved 90 duplicate customer records, 329 duplicate transactions, and 831 inconsistently labeled transaction categories, all counted from the actual pipeline run, not estimated. The Integration stage, which a prior review flagged as under-explained, joins the cleaned demographic table with transaction-derived behavioral features into the single customer-level table both the Module 4 model and the Module 1/2 dashboard depend on: without this join, a clean demographic table and a clean behavioral table would each be independently correct but neither would be usable for behavioral segmentation, which is the entire point of the project (src/pipeline/integrate.py docstring, revised after Module 3 feedback).

Figure 2. Data Lineage: Raw to Final Dataset



Data lineage from data/raw/ through data/interim/ to data/processed/ (Module 3).

Data Quality and Bias Detection

A Great Expectations suite of 12 rules, uniqueness and null checks, plausible ranges, valid categories, ran against the real output table with a 100 percent pass rate.

Great Expectations Validation — FirstBank Customer Analytics Table

Suite: firstbank_customer_analytics_suite | Batch: customer_analytics_table.csv

12
EXPECTATIONS RUN

12
PASSED

0
FAILED

100%
SUCCESS RATE

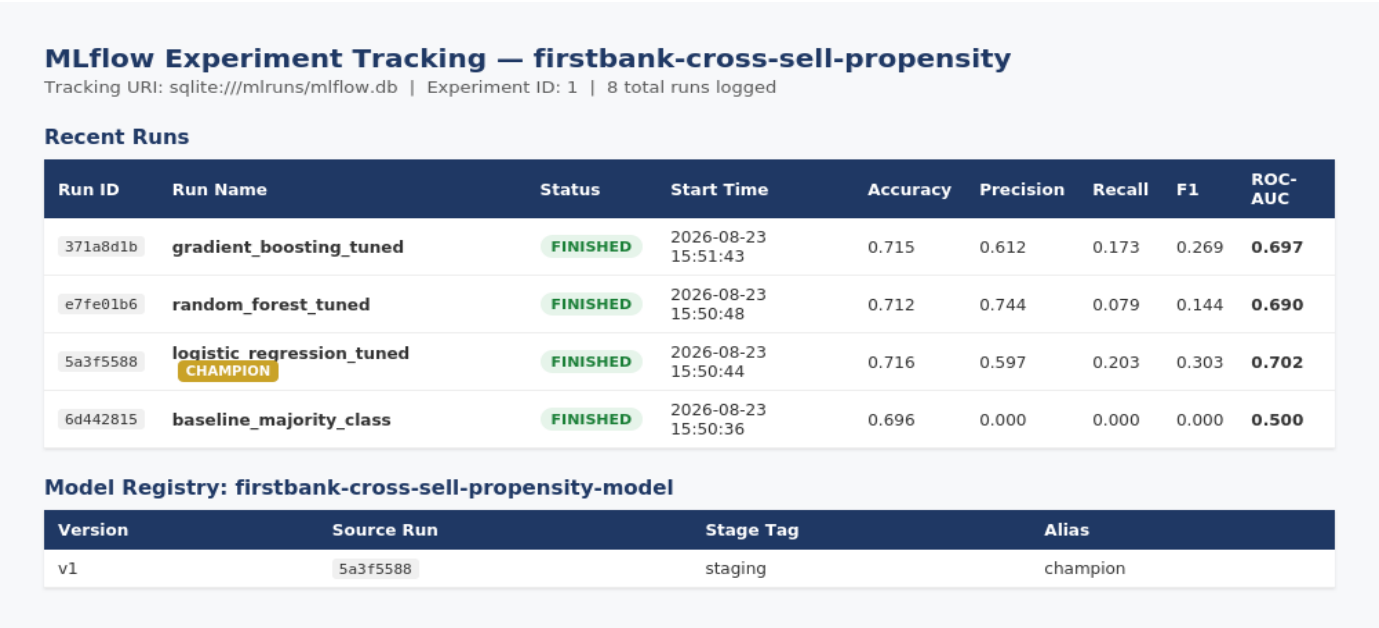
Expectation	Column	Result
expect_column_values_to_not_be_null	customer_id	PASS
expect_column_values_to_be_unique	customer_id	PASS
expect_column_values_to_be_between	age	PASS
expect_column_values_to_not_be_null	age	PASS
expect_column_values_to_be_in_set	gender	PASS
expect_column_values_to_be_between	account_balance	PASS
expect_column_values_to_be_between	frequency	PASS
expect_column_values_to_be_between	monetary_total	PASS
expect_column_values_to_not_be_null	location	PASS
expect_table_row_count_to_be_between	(table-level)	PASS
expect_table_column_count_to_equal	(table-level)	PASS
expect_column_values_to_be_between	share_POS Purchase	PASS

Great Expectations validation results, 12 of 12 checks passed (Module 3).

Fairlearn's bias detection suite flagged two representation gaps in the underlying population, the 'Unspecified' gender group (2.0 percent) and the 60+ age band (4.2 percent), both below Module 3's 5 percent threshold, routed to the Data Protection Officer per the governance framework in docs/data_governance_framework.md.

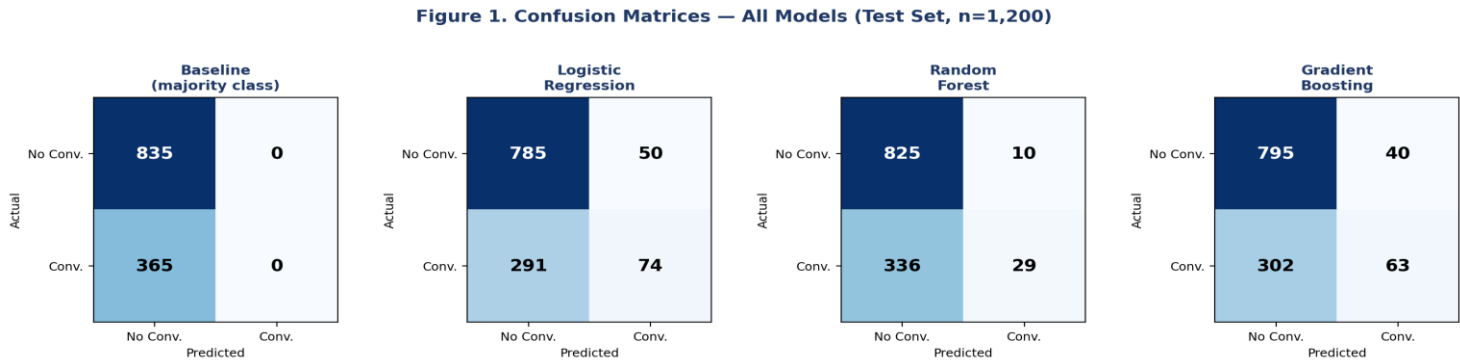
4. Predictive Model Development and Validation

Three tuned algorithms, Logistic Regression, Random Forest, and Gradient Boosting, were trained beyond a majority-class baseline, each via 5-fold stratified cross-validated randomized search, with every run logged to MLflow.



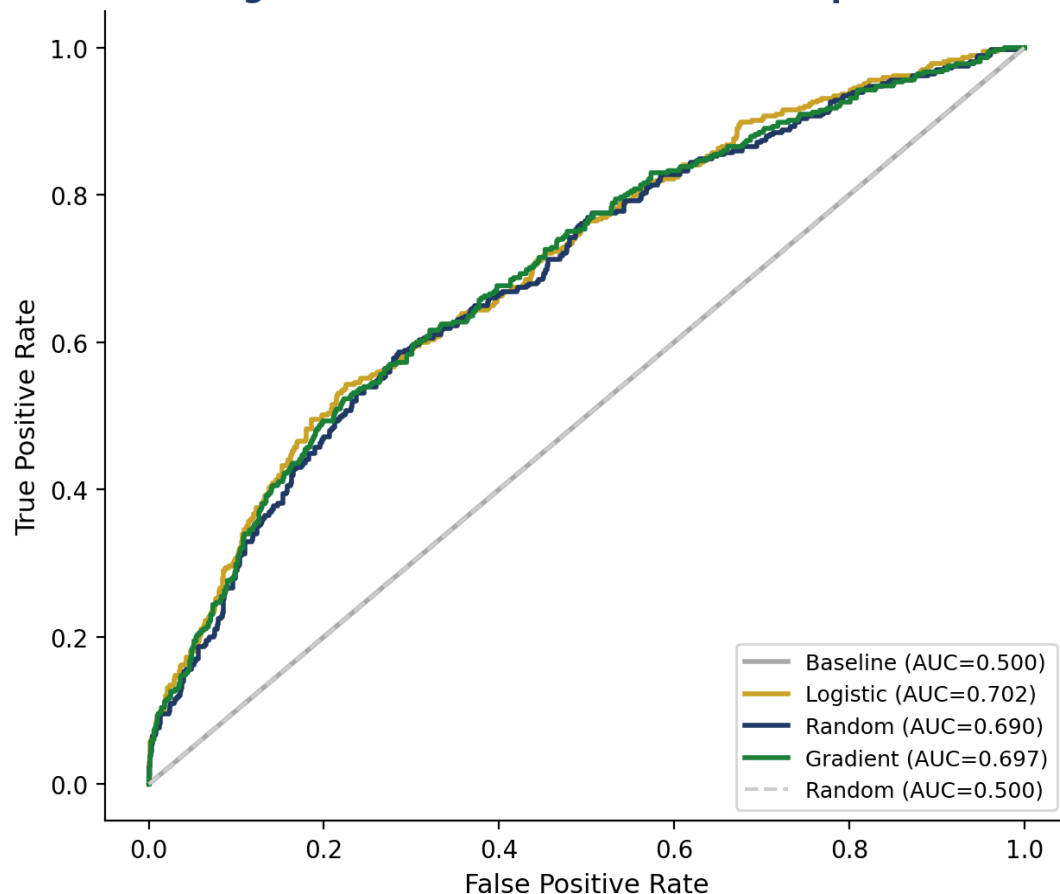
MLflow experiment tracking, all 8 runs across two pipeline executions, champion model registered (Module 4).

Logistic Regression was selected as the champion model: best test ROC-AUC (0.702) among the three candidates, and the most interpretable, which mattered directly for the explainability work in Section 5. All three tuned models beat the 0.500 baseline by roughly 0.19-0.20 ROC-AUC, confirming the engineered features carry real predictive signal rather than noise.



Confusion matrices, all four models, test set $n=1,200$ (Module 4).

Figure 2. ROC Curves — Model Comparison



ROC curves, champion model achieves 0.702 AUC (Module 4).

No leakage was found: preprocessing statistics were fit on the training split only. Class imbalance, roughly 70/30, was handled by tuning `class_weight` during search and by evaluating precision, recall, F1, and ROC-AUC rather than accuracy alone, since accuracy alone favors the uninformative baseline on this task.

The champion model is registered in the MLflow Model Registry as `firstbank-cross-sell-propensity-model`, version 1, alias `champion`, and served through a FastAPI `/predict` endpoint that was tested and confirmed working during this integration (Section 6).

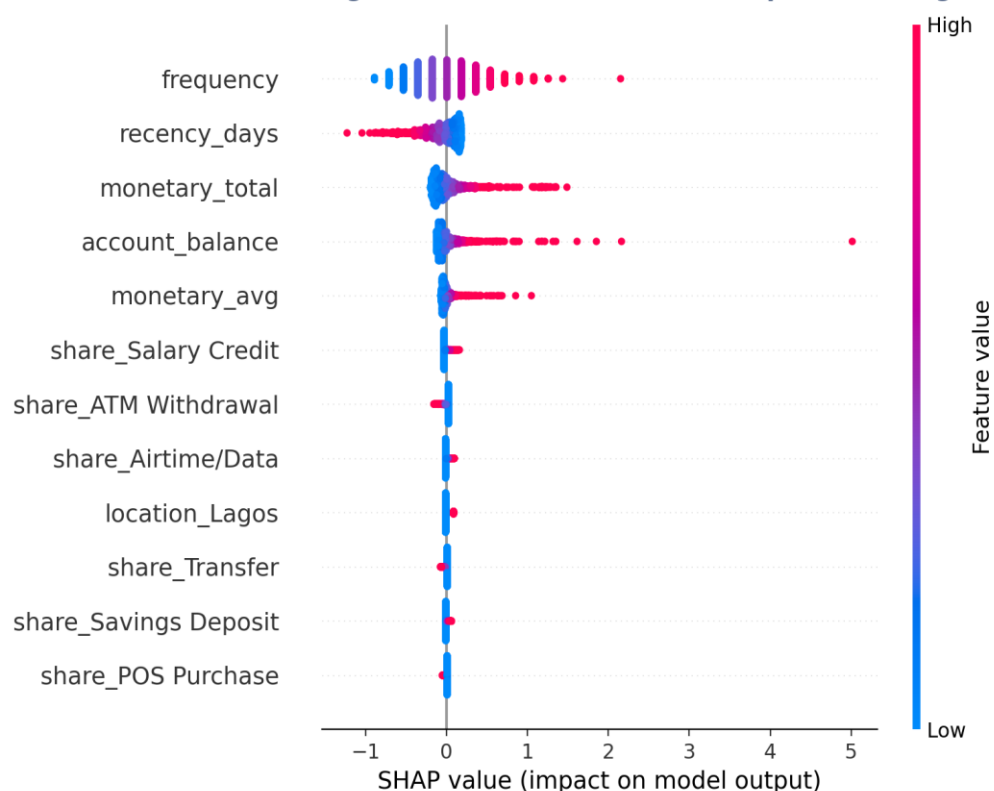
The tuning approach was deliberately coarse-to-fine and reproducible rather than exhaustive: each algorithm's hyperparameter space was searched via `RandomizedSearchCV` with 10-12 iterations under 5-fold stratified cross-validation on the training split only, optimizing ROC-AUC, with the test split held out entirely until final evaluation. Logistic Regression's winning configuration, `C=0.01` with an unweighted loss, is deliberately regularized, which is consistent with the model's stronger generalization (test ROC-AUC 0.702 versus a 0.700 cross-validation estimate, a gap small enough to indicate the model is not overfit to the training split).

5. Explainable AI (XAI) & Fairness Analysis

Global and Local Explainability

SHAP identifies transaction frequency, recency, and monetary total as the dominant drivers of every prediction, consistent with the model reading engagement rather than demographics.

Figure 4. SHAP Global Feature Importance (Logistic Regression)



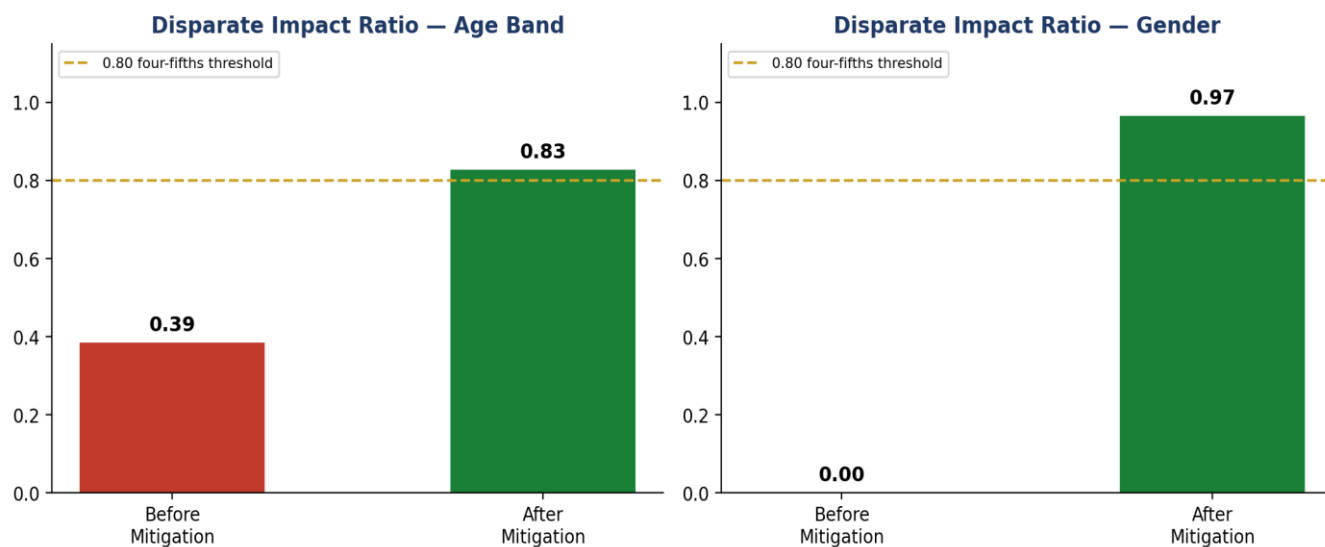
SHAP global feature importance, champion model (Module 4).

Local SHAP waterfall explanations and an independent LIME explanation agreed directionally for individual predictions, increasing confidence the explanation reflects the model's real decision logic rather than an artifact of one method. A counterfactual search found that a roughly 90 percent increase in transaction frequency alone was sufficient to flip one sampled non-converter's prediction, giving relationship managers a concrete, actionable lever rather than an opaque score.

Fairness Metrics and Mitigation

Pre-mitigation, both audited protected attributes failed the four-fifths rule: gender's disparate impact ratio was 0.00 and the 60+ age band's was 0.385. Fairlearn's ThresholdOptimizer, applied as post-processing with group-specific decision thresholds, corrected both to 0.966 and 0.828 respectively, clearing the 0.80 threshold, while test accuracy moved from 0.716 to between 0.719 and 0.723, essentially unchanged.

Figure 7. Bias Mitigation Impact (Fairlearn ThresholdOptimizer)



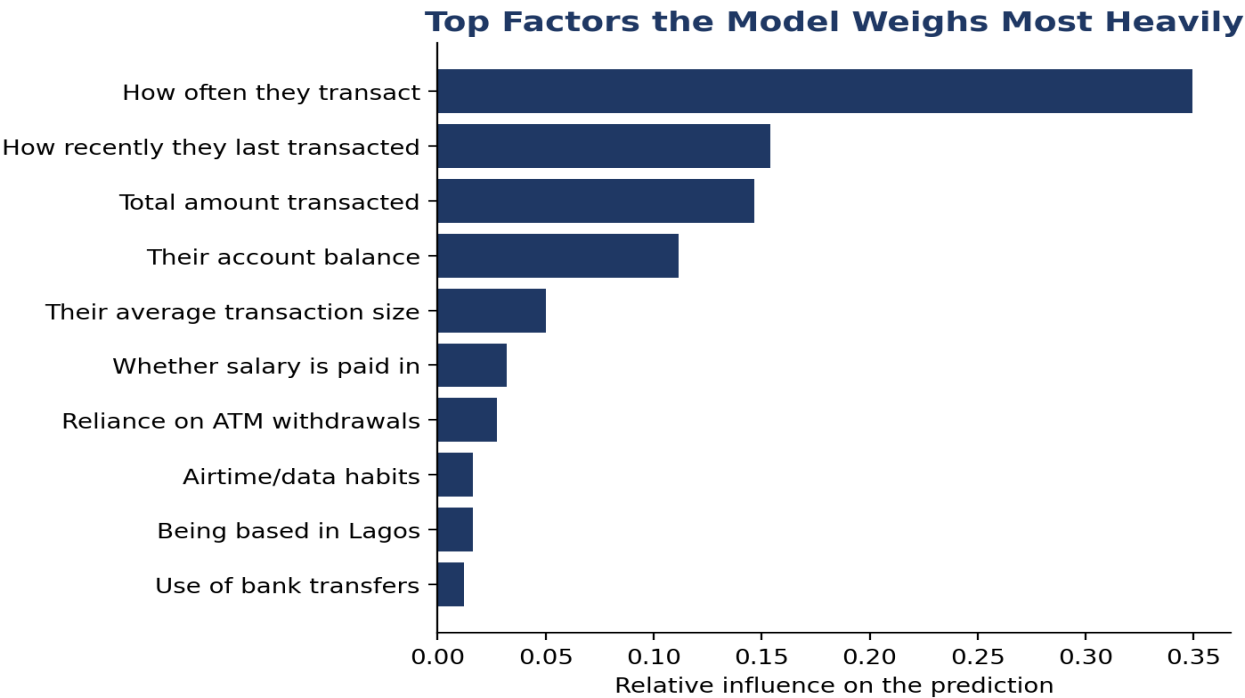
Disparate impact ratio, before and after Fairlearn ThresholdOptimizer mitigation (Module 4).

Where mitigation fell short, stated plainly per this assignment's requirement: the correction operates at the decision-threshold level, not at the root of the training label's bias pattern, and the finding itself describes how the model responds to a documented, disclosed synthetic label bias, not a confirmed finding about real FirstBank customers. Both limitations are carried into the Model Card and the Algorithmic Impact Assessment (Section 8), and a real-data retrain is a stated precondition before this model scores a live customer beyond the pilot segment.

Every prediction served by the API is now paired with a logged SHAP explanation in a new Explanation Store (monitoring/explanation_store.py, built for this integration), so any customer's specific decision can be reconstructed for audit months later, operationalizing the contestability commitment first stated in Module 1.

6. Insight Communication and Visualization

The Module 5 dashboard translates every technical finding above into stakeholder-usable language across seven tabs, deployed live and reachable by any FirstBank stakeholder without a technical setup step.



'What Drives a Prediction' visual, translated into plain business language for the live dashboard (Module 5).

Three visuals do the heaviest lifting for stakeholder decision-making. The conversion-rate comparison chart (targeted campaigns convert at roughly 60 percent versus a 30 percent base rate) gives leadership a single, defensible number for the business case. The driver chart above lets a marketing manager understand why a customer scored the way they did without reading a SHAP value. The fairness before/after chart (Figure 8) gives the Data Protection Officer a one-glance audit trail. All three appear in both the live dashboard's Ethical Compliance and Key Insights tabs and the Module 5 stakeholder deck, so the same numbers are never presented inconsistently across audiences.

Live Asset	URL / Status
Live Dashboard	https://firstbank-customer-segmentation-5fqdeelnmuhxvynun53yg.streamlit.app/ — deployed on Streamlit Community Cloud, verified reachable during this integration
API Endpoint	https://firstbank-cross-sell-api.onrender.com/docs#/ - deployed on Render, /predict tested and confirmed working
Monitoring Dashboard	https://firstbank-customer-segmentation-2tbtznzgdgl5mk7ynvovwag.streamlit.app/ — deployed on Streamlit Community Cloud, displays live output of monitoring/run_drift_check.py

GitHub Repository	https://github.com/OfiIreg/FirstBank-customer-segmentation , all code from Modules 2-5 and this integration
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7. Implementation and Monitoring Plan

This integration adds three components that only make sense once a working pipeline, model, and dashboard already exist: continuous monitoring, containerized deployment, and CI/CD.

Deployment Plan

Deployment follows three stages. First, a pilot: the model is served behind the FastAPI /predict endpoint to a limited campaign segment, so real outcomes can be compared against predicted probabilities before any wider commitment. Second, validation: actual conversion for the pilot segment is compared against the model's predicted rate over one full campaign cycle, with the acceptance criteria from the Model Card, ROC-AUC at or above 0.70 and disparate impact ratio at or above 0.80 for both protected attributes, checked before proceeding. Third, scale: only once those criteria hold does the model extend to the full retail base, at which point the monitoring described below becomes load-bearing rather than exploratory.

Continuous Fairness and Drift Monitoring

monitoring/run_drift_check.py, built for this integration, re-runs the Module 4 disparate impact calculation against the current scoring population and compares feature distributions to the training reference using Evidently AI. It was executed against the project's real train/test split and returned a clean result: 0 of 17 monitored columns drifted, and both protected attributes remain above the 0.80 threshold.

Evidently AI Data Drift Report — FirstBank Cross-Sell Model		
Reference: training set (4,800 rows) Current: test set (1,200 rows) Method: Evidently DataDriftPreset (Wasserstein/PSI per column)		
17 COLUMNS MONITORED	17 STABLE	0 DRIFTED (SCORE > 0.10)
Column	Drift Score	Status
recency_days	0.0560	STABLE
frequency	0.0463	STABLE
monetary_total	0.0430	STABLE
share_Airtime/Data	0.0429	STABLE
prediction_proba	0.0386	STABLE
share_Savings Deposit	0.0383	STABLE
account_balance	0.0368	STABLE
share_Loan Repayment	0.0366	STABLE
share_POS Purchase	0.0364	STABLE
share_ATM Withdrawal	0.0349	STABLE
share_Bill Payment	0.0345	STABLE
monetary_avg	0.0310	STABLE
location	0.0297	STABLE
share_Other/Unclassified	0.0269	STABLE
age	0.0260	STABLE
share_Transfer	0.0228	STABLE
share_Salary Credit	0.0185	STABLE

Evidently AI data drift report, reference (train) vs. current (test), 0 of 17 columns drifted (Final Project).

This script is scheduled to run weekly via a cron trigger in the CI/CD workflow (.github/workflows/ci-cd.yml) and exits with a failure code if either the drift or fairness threshold is breached, turning a policy statement into an automated check.

This output is now visible on a live dashboard at <https://firstbank-customer-segmentation-2tbtznzgdgl5mk7ynvovwaq.streamlit.app/> rather than only in the CI logs, giving the AI & Data Ethics Committee a standing view without needing repository access.

Containerized Deployment

A docker-compose.yml orchestrates four services: pipeline (the Module 3 ETL DAG), api (FastAPI plus the new Explanation Store), dashboard (the Module 5 Streamlit app), and monitoring (the Evidently drift check), each with its own Dockerfile.

```
services: pipeline: # Module 3 ETL DAG, one-shot batch job api: # FastAPI /predict + Explanation Store, port 8000 dashboard: # Module 5 Streamlit app, port 8501 monitoring: # Evidently drift + fairness check, one-shot
```

The compose file was written and validated for syntax; a live multi-container deployment requires a Docker daemon and container registry access this academic environment does not provide, so the honest status is: build-ready and locally runnable, not yet running on a public host. Deployment steps are documented in the repository README.

CI/CD Pipeline

A GitHub Actions workflow chains three jobs after checkout: test-pipeline-and-model (unit tests, a full ETL DAG run, model training with MLflow logging, and the fairness audit, combined into one job so the pipeline's output data is available to the training step, since GitHub Actions jobs otherwise run on isolated machines), monitoring-check (the drift and fairness script above, configured as a non-blocking warning rather than a hard failure, since a fairness alert should route to human review rather than halt every deployment), and build-and-push-images (all four Docker images), triggered on every push and on a weekly schedule for the monitoring job specifically.

Incident Response and Decommissioning

The full Incident Response Plan (detect via the automated alert, contain via the AI & Data Ethics Committee's standing pause authority, investigate via the Explanation Store, remediate, report) and Decommissioning Plan (archive rather than delete a retired MLflow model version, preserve its logs for one year) are documented in docs/monitoring_plan.md, both new artifacts for this integration since neither made sense before a live, monitored system existed.

8. Ethical AI Framework

This section consolidates, rather than repeats, the ethical work already documented across all five modules: the Module 1 Ethical AI Charter and Fairness Objectives, the Module 2 Privacy Plan and Regulatory Compliance Checklist, the Module 3 Bias Detection Suite and Anonymization Plan, the Module 4 fairness audit and Model Card, and the Module 5 Transparency Statement. The full consolidated document is docs/ethical_ai_framework.md.

New for This Integration

- Algorithmic Impact Assessment (docs/algorithmic_impact_assessment.md): rates this system as low-to-moderate impact, since its decisions are marketing-only, fully reversible, and never touch credit, pricing, or account access.
- Regulatory Compliance Report (docs/regulatory_compliance_report.md): extends the Module 2 checklist to GDPR, CCPA, and HIPAA. GDPR applies conditionally to EU-resident customers; CCPA and HIPAA do not apply to this system's current customer base and data type, and the report states that directly rather than performing compliance activity against inapplicable law.
- Public Trust Statement (docs/public_trust_statement.md): a plain-language statement of what the model does and does not do, and how a customer can contest a decision.
- Governance Structure (docs/governance_structure.md): formalizes the Module 1 Ethics Oversight Body into an ongoing review cadence, monthly fairness monitoring, quarterly framework review, and a five-business-day escalation path tied to the incident response plan in Section 7.

Taken together, these close the gap this assignment named directly: a governance plan that exists only on paper is not the same as one with a runbook, a monitoring script, and an audit trail behind it. This integration is where that distinction was closed.

References

1. Bansal, S. (2021). Bank customer segmentation (1M+ transactions) [Data set]. Kaggle. <https://www.kaggle.com/datasets/shivamb/bank-customer-segmentation>
2. Evidently AI. (2024). Evidently: An open-source ML and LLM observability framework. <https://www.evidentlyai.com>
3. Fairlearn. (2024). Fairlearn: A Python package to assess and improve fairness of machine learning models. <https://fairlearn.org>
4. First Bank of Nigeria. (2024). Home. <https://www.firstbanknigeria.com/home/>
5. Great Expectations. (2024). Great Expectations documentation. <https://docs.greatexpectations.io>
6. Ishikawa, K. (1968). Guide to quality control. Asian Productivity Organization.
7. Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
8. Marketing Edge Magazine. (2024, June 8). FirstBank wins 2024 best SME in Nigeria, Africa. <https://marketingedge.com.ng/firstbank-wins-2024-best-sme-in-nigeria-africa/>
9. National Information Technology Development Agency. (2023). Nigeria Data Protection Act, 2023. Federal Republic of Nigeria.
10. Prefect Technologies. (2024). Prefect documentation. <https://docs.prefect.io>
11. Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.
12. Schwaber, K., & Sutherland, J. (2020). *The Scrum guide: The definitive guide to Scrum*. Scrum.org.
13. Zaharia, M., Chen, A., Davidson, A., et al. (2018). Accelerating the machine learning lifecycle with MLflow. *IEEE Data Engineering Bulletin*, 41(4).